FARM MANAGEMENT SYSTEM

PROJECT DOCUMENTATION

A Web-Based Application for Efficient Farm Planning and Management

Submitted in partial fulfillment of the requirements for the degree of

B.Sc. Computer Science

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CERTIFICATE

This is to certify that the project entitled “Farm Management System” is a bonafide work carried out by _____, a student of B.Sc. Computer Science, during the academic year 2026–2027.

The project has been developed under the guidance and supervision of the faculty members of the Department of Computer Science. The work presented in this documentation is submitted in partial fulfillment of the requirements for the award of the degree.

Guide Signature: _____ Head of Department: _____

Date: _____ Place: _____

DECLARATION

I hereby declare that the project entitled “Farm Management System” is my original work carried out as part of my academic project. The information and implementation presented in this report are prepared for educational purposes.

I further declare that this project has not been submitted in full or in part to any other institution for the award of any degree or qualification.

Student Signature: _____

Name: _____

Date: _____

ACKNOWLEDGEMENT

I express my sincere gratitude to the management of the institution and the Department of Computer Science for providing the opportunity and facilities required to complete this project.

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ABSTRACT

The Farm Management System is a web-based application designed to help farmers organize and manage important farm activities in a simple and systematic manner. Traditional farm management often depends on notebooks, registers and manual calculations, which may lead to data loss, duplication and difficulty in tracking activities.

The proposed system provides modules for farmer details, land management, crop management, inventory, farming activities, expenses, sales and reports. The system stores information in a centralized database so that users can retrieve and update records easily.

The application can reduce paperwork, improve record accuracy and support better planning. It can also provide summarized information about farm operations and expenses, helping the user understand the current status of the farm.

TABLE OF CONTENTS

1. Introduction 7

2. Existing System 8

3. Proposed System 9

4. Objectives 10

5. Scope and Feasibility 11

6. Requirements 12

7. System Analysis 13

8. System Design 14

9. Modules 15

10. Database Design 16

11. Data Flow Description 17

12. User Interface Design 18

13. Implementation 19

14. Testing 20

15. Test Cases 21

16. Security and Validation 22

17. Advantages and Limitations 23

18. Future Enhancements 24

19. Conclusion and Bibliography 25

1. INTRODUCTION

1.1 Background

Agriculture involves several activities such as land preparation, seed selection, crop cultivation, irrigation, fertilizer application, pest control, harvesting, storage and sales. Managing these activities manually can be difficult when the farm has many crops or repeated activities.

1.2 Project Overview

The Farm Management System provides a centralized platform to record and manage farm-related information. The system is intended to make routine record keeping easier and more organized.

1.3 Need for the System

A digital system can help maintain consistent records, reduce manual calculations and make information available whenever required. It also provides a foundation for generating reports about crops, expenses and farm activities.

2. EXISTING SYSTEM

2.1 Manual Method

In many small-scale farming environments, information is maintained in notebooks, paper registers or separate files. Details about crops, expenses, labour, stock and sales may be recorded independently.

2.2 Problems

Manual records can be misplaced or damaged. Searching for older information can take time. Calculations of expenses and sales may contain errors, and it may be difficult to get a quick summary of the farm.

2.3 Drawbacks

The existing approach has limited data organization, limited reporting capability, greater dependency on physical records and difficulty in maintaining historical information.

3. PROPOSED SYSTEM

3.1 Overview

The proposed Farm Management System is a database-driven web application. It allows authorized users to enter, update, view and manage farm information through simple web pages.

3.2 Main Features

The system includes farmer management, land records, crop records, activity tracking, inventory management, expense tracking, sales records and report generation.

3.3 Benefits

Centralized storage improves organization and retrieval. The system also supports systematic monitoring of activities and provides summarized information for day-to-day management.

4. OBJECTIVES

The major objectives of the Farm Management System are:

• To maintain farmer and farm details in a structured database.

• To record land and crop information accurately.

• To track farming activities and their dates.

• To manage seeds, fertilizers, pesticides and other inventory items.

• To record farm expenses and sales transactions.

• To reduce paperwork and manual record keeping.

• To provide useful reports for monitoring farm operations.

• To provide a simple and user-friendly interface.

5. SCOPE AND FEASIBILITY

5.1 Scope

The system can be used by individual farmers, small farm owners and agricultural organizations for maintaining operational records. It can be expanded later with mobile access, weather information and analytics.

5.2 Technical Feasibility

The application can be implemented using common web technologies such as HTML, CSS, PHP and MySQL. It can run on a local server using XAMPP and can also be deployed on a suitable web server.

5.3 Economic Feasibility

The proposed system can be developed using open-source software and commonly available hardware, making the basic implementation cost-effective.

5.4 Operational Feasibility

The interface is designed for simple data entry and viewing. Basic computer knowledge is sufficient for operating the system.

6. REQUIREMENTS

6.1 Hardware Requirements

Computer or laptop, minimum 4 GB RAM, sufficient storage space, keyboard, mouse and display. A printer is optional for reports.

6.2 Software Requirements

Operating system such as Windows or Linux; web browser such as Chrome or Edge; XAMPP for Apache and MySQL; Visual Studio Code or another code editor.

6.3 Technologies

Frontend: HTML and CSS. Backend: PHP. Database: MySQL. Development environment: XAMPP and Visual Studio Code.

6.4 Functional Requirements

The system shall allow login, adding and updating farm information, managing crops and inventory, recording activities, recording expenses and sales, and generating reports.

6.5 Non-Functional Requirements

The system should provide usability, reasonable response time, data consistency, maintainability and basic security.

7. SYSTEM ANALYSIS

7.1 User Analysis

The primary user is the farmer or farm administrator. The user enters farm data and uses reports to monitor activities.

7.2 Input Analysis

Typical inputs include farmer name, land details, crop name, crop variety, sowing date, harvest date, quantity, inventory quantity, expense amount and sales amount.

7.3 Process Analysis

The system validates input, stores records in MySQL and retrieves information for display and reports.

7.4 Output Analysis

Outputs include crop lists, activity history, inventory status, expense summaries, sales records and overall farm reports.

8. SYSTEM DESIGN

8.1 Architecture

The system follows a simple three-layer approach: presentation layer using HTML/CSS, application layer using PHP, and data layer using MySQL.

8.2 Processing Flow

User Login → Dashboard → Select Module → Enter/Update Data → Validate Data → Store in Database → Display Confirmation/Report.

8.3 Design Principles

The system uses modular design so that each major function can be developed and maintained separately. Database tables are separated according to entities to reduce duplication.

9. MODULES

9.1 Login Module

Provides authentication for authorized users.

9.2 Farmer/Farm Module

Stores farmer and farm profile information.

9.3 Land Module

Maintains land location, area and soil-related information.

9.4 Crop Module

Stores crop name, variety, sowing date, expected harvest date and status.

9.5 Activity Module

Records irrigation, fertilizer application, pest control, harvesting and other activities.

9.6 Inventory Module

Tracks seeds, fertilizers, pesticides, tools and available quantities.

9.7 Expense Module

Stores farm-related expenses and categories.

9.8 Sales Module

Records harvested products sold, quantity, rate and total amount.

9.9 Reports Module

Displays summarized information from the stored records.

10. DATABASE DESIGN

The database is designed to store information in related tables. A sample structure is given below.

Table	Important Fields
users	user_id, username, password
farm	farm_id, farmer_name, location, total_area
land	land_id, farm_id, area, soil_type
crop	crop_id, land_id, crop_name, variety, sowing_date, harvest_date
activity	activity_id, crop_id, activity_type, activity_date, description
inventory	item_id, item_name, category, quantity, unit
expense	expense_id, category, amount, expense_date, description
sales	sale_id, crop_id, quantity, rate, total_amount, sale_date

Primary and foreign keys are used to establish relationships between farm, land, crop and transaction records.

11. DATA FLOW DESCRIPTION

11.1 Context Level

The farmer/user interacts with the Farm Management System by entering data and requesting information. The system processes the request and returns confirmations or reports.

11.2 Level 1 Flow

User → Login → Dashboard → Farm/Land/Crop/Activity/Inventory/Expense/Sales modules → MySQL Database → Reports/Results.

11.3 Data Validation

Required fields should be checked before insertion. Numeric fields should accept valid numeric values, dates should follow the expected format and related IDs should refer to existing records.

Suggested DFD documentation placeholder: Insert Level-0 and Level-1 DFD diagrams here in the final college report.

12. USER INTERFACE DESIGN

12.1 Login Page

Contains username and password fields with a login button.

12.2 Dashboard

Provides navigation to farm, land, crop, activity, inventory, expense, sales and reports modules.

12.3 Data Entry Pages

Forms contain labels, input fields, date selectors, dropdowns and action buttons such as Save, Update and Delete.

12.4 Reports Page

Displays stored records in tables and provides summarized information.

Screenshot placeholders for final documentation:

• Figure 1 – Login Page

• Figure 2 – Dashboard

• Figure 3 – Crop Management

• Figure 4 – Inventory Management

• Figure 5 – Expense/Sales Report

13. IMPLEMENTATION

13.1 Frontend

HTML is used to create page structure and forms. CSS is used for layout, readability, navigation and responsive presentation.

13.2 Backend

PHP handles form submission, validation, database connectivity, CRUD operations and report generation.

13.3 Database

MySQL stores user, farm, crop, activity, inventory, expense and sales information.

13.4 Development Process

Create database → create tables → connect PHP to MySQL → develop login → create modules → implement CRUD operations → test each module → integrate the system.

13.5 CRUD Operations

Create adds a new record, Read displays records, Update modifies an existing record and Delete removes a record after appropriate confirmation.

14. TESTING

14.1 Purpose

Testing verifies that the application works according to the specified requirements and that incorrect input is handled appropriately.

14.2 Unit Testing

Each module is tested independently, such as crop insertion, inventory update and expense calculation.

14.3 Integration Testing

Modules are tested together to ensure that related records and database relationships work correctly.

14.4 System Testing

The complete application is tested from login through report generation.

14.5 User Acceptance Testing

The final interface is checked for ease of use and whether it supports expected farm management tasks.

15. TEST CASES

ID	Test	Expected Result
TC01	Valid login	Dashboard opens
TC02	Invalid login	Error message shown
TC03	Add crop	Crop saved in database
TC04	Edit crop	Updated data displayed
TC05	Delete record	Record removed after confirmation
TC06	Add inventory	Quantity stored correctly
TC07	Add expense	Expense stored and displayed
TC08	Add sale	Sale total calculated/stored
TC09	Missing required field	Validation message shown
TC10	Generate report	Correct records displayed

The test cases should be executed during the final implementation and the actual result/status can be added to the project record.

16. SECURITY AND VALIDATION

16.1 Authentication

Only authenticated users should access protected pages.

16.2 Input Validation

Form inputs should be checked for empty values, valid data types, valid dates and reasonable numeric values.

16.3 Database Security

Database credentials should not be exposed in the user interface. Passwords should be stored securely using suitable password hashing in a production system.

16.4 Session Management

PHP sessions can be used to maintain authenticated user state and restrict direct access to internal pages.

16.5 Backup

Regular database backups should be maintained to reduce the risk of data loss.

17. ADVANTAGES AND LIMITATIONS

17.1 Advantages

Centralized record keeping; reduced paperwork; faster searching; organized crop and activity records; easier expense and sales tracking; improved report generation; expandable architecture.

17.2 Limitations

The basic version depends on accurate data entry. Internet access may be required if deployed online. Weather, market price and external agricultural information are not automatically included unless additional services are integrated.

17.3 Practical Consideration

The system is intended as a management and record-keeping tool. It does not replace professional agricultural advice or field-level decision making.

18. FUTURE ENHANCEMENTS

The Farm Management System can be extended with the following features:

- Mobile application for Android devices.
- Weather forecast and rainfall information integration.
- Crop disease identification using image-based AI.
- Market price and crop demand information.
- SMS/email notifications for important activities.
- Fertilizer and irrigation reminders.
- Graphs and dashboards for expenses, yield and sales.
- Multi-farm and multi-user support.
- Cloud database and secure online backup.
- Regional language support for easier adoption.

19. CONCLUSION AND BIBLIOGRAPHY

19.1 Conclusion

The Farm Management System provides a structured approach for maintaining important farm records. By replacing scattered manual records with a centralized database application, the system can make information easier to store, retrieve and monitor. The modular design also allows additional agricultural and analytical features to be incorporated in the future.

19.2 Bibliography

1. PHP Documentation – php.net
2. MySQL Documentation – dev.mysql.com
3. HTML and CSS learning resources – developer.mozilla.org
4. XAMPP Documentation – apachefriends.org
5. General references on database management systems and software engineering.

19.3 Project Outcome

The completed application is expected to demonstrate practical knowledge of web development, database management, CRUD operations, system analysis, testing and documentation.